Zijie Jin

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EDUCATION

University of Washington **Sep 2023 - Present**
PhD student – Electrical and Computer Engineering
Advisor: Kimberly Ingraham

McGill University **Sep 2019 – Apr 2022**
Master of Engineering – Biological & Biomedical Engineering

- BBME Excellence Award 2019 – 2021

McGill University **Sep 2014 - Dec 2018**
Bachelor of Engineering – Chemical Engineering, Minor in Biotechnology
Graduation with Dean's Honour List

- ACDEG Scholarship in Chemical Engineering 2015 - 2016
- Dr. P.K. Ganguli Memorial Scholarship 2016 - 2017
- Louis C Ho Scholarship in Engineering 2018 - 2019
- Dean's Honours List 2015, 2016 and 2018
- Tomlinson Engagement Award for Mentoring (TEAM) 2018

RESEARCH EXPERIENCES

Ingraham Lab
University of Washington
PhD student **Sep 2023 – Present**

- Investigating and modelling, using a game theory paradigm, how humans interact and co-adapt with assistive wearable devices in the context of locomotion biomechanics
- Developing adaptive control schemes for lower-limb exoskeletons to guide human gait for achieving distinctive goals (such as minimizing metabolic cost)

Laboratoire de Conception de Systèmes Intelligents et Mécatroniques (CoSIM)
Polytechnique Montreal
Research Assistant – Biomechanics and exoskeleton **Aug 2022 – April 2023**

- Modeling of dynamics of human lower limb wearing 1-DOF knee exoskeleton and human sit-to-stand motion wearing 3-DOF lower limb exoskeleton
- Simulating the control strategy of exoskeletons implementing sliding mode controllers with non-linear disturbance observer for human joint torque

Micro and Nano Bioengineering Lab
McGill University
Master's Student **Sep 2019 – Apr 2022**
Mini Colon Model (MiCoMo) Project

- Developed a low-cost *in vitro* artificial colon simulator that mimics fundamental features of the human GI microbial environment with small footprint, quick stabilization time and high throughput
- Programmed microcontroller for automatic control of physiological parameters, including anaerobic condition, temperature, mixing and pH
- Validated the device with 16S rRNA sequencing to examine bacterial diversity, taxonomy and stabilization in community

COVID-19 RDT Project

- Participated in a team effort to develop a Point-Of-Care (POC) device for diagnosis of antibody against COVID19 based on lateral flow strips and 3D printed chip with lab-developed technology, Domino Capillaric Circuits.
- Tested and optimized reagent concentration and volume for establishment of lateral flow strips with high sensitivity
- Developed a cell phone app and auxiliary setup for imaging the lateral flow strip for quick analysis and quantification of signal

EV investigation with Microarray Project

- Participated in a project to develop a method for investigating cell extracellular vesicles (EV) subpopulation with protein microarray and normalizing fluorescence signal with universal dye and universal surface target to remove EV population size bias and reveal new information regarding EV subpopulation
- Conducted microarray data extraction, signal-processing, analysis and mathematical interpretation

PUBLICATIONS AND PRESENTATIONS

Journal Publications

- Mohamed Yafia, Oriol Ymbern, Ayokunle O. Olanrewaju, Azim Parandakh, Ahmad Sohrabi Kashani, Johan Renault, **Zijie Jin**, Geunyong Kim, Andy Ng & David Juncker. *Microfluidic chain reaction of structurally programmed capillary flow events. Nature* **605**, 464–469 (2022). https://www.nature.com/articles/s41586-022-04683-4

- **Zijie Jin**, Andy Ng, Corinne F. Maurice and David Juncker. *The Mini Colon Model: a benchtop multi-bioreactor system to investigate the gut microbiome. Gut Microbes* 14:1 (2022) https://doi.org/10.1080/19490976.2022.2096993

Manuscript Preprints

- Molly L. Shen, Andreas Wallucks*, **Zijie Jin*,** Rosalie Martel*, Lucile Alexandre, Philippe DeCorwin-Martin, Lorenna Oliveira Fernandes de Araujo, Andy Ng, Peter M. Siegel, David Juncker. *Co-Enrichment of Proteins in Extracellular Vesicles*
  https://arxiv.org/abs/2301.04051

- **Zijie Jin,** Jason Isa, Samuel A Burden, Kimberly A Ingraham. *Quantifying human adaptation to a novel split-belt walking condition after broad experience at different belt speeds*
  https://www.biorxiv.org/content/10.1101/2025.08.25.671956v1.full.pdf

Oral Presentations

- **Zijie Jin**, Andy Ng, Corinne F. Maurice, & David Juncker. *MiCoMo: a miniature multi-bioreactor system for stable and reproducible in vitro investigation of gut microbiome.* The 8th World Congress on Targeting Microbiota (2021)

Poster Presentations

- **Zijie Jin**, Jason T. Isa, Samuel A. Burden, and Kimberly A. Ingraham. *Influencing human gait mechanics with adaptive split-belt treadmill*. American Society of Biomechanics Meeting 2024 (ASB 2024)

- **Zijie Jin**, Jason T. Isa, Samuel A. Burden, and Kimberly A. Ingraham. *Influencing human gait mechanics with adaptive split-belt treadmill*. Northwest Biomechanics Symposium (NWBS 2024)

- **Zijie Jin**, Andy Ng, Corinne F. Maurice, & David Juncker. *MiCoMo: a miniature multi-bioreactor system for investigation of gut microbiome.* The 25th International Conference on Miniaturized Systems for Chemistry and Life Sciences
  (μTAS 2021)

- **Zijie Jin**, Will Jogia, Andy Ng, Corinne F. Maurice, & David Juncker, D. *GITbox: a miniature multi-bioreactor system for stable and reproducible in vitro investigation of gut microbiome* The 6th World Congress of the Tissue Engineering and

Regenerative Medicine International Society (TERMIS 2021)

Teaching Experiences and Services

University of Washington, Department of Electrical & Computer Engineering **Sep 2025 - Present**
Future Faculty Preparation Program (F2P2) Committee member

• Organized F2P2 events with other committee members to help current PhD students aspiring for academic careers, including CV review workshops for faculty application and podcast/panel discussions with students currently in job market and new professors.

McGill University, Department of Chemical Engineering **Sep 2018 - Dec 2018**
Class Mentor for CHEE 474

• Worked as class mentor for course “CHEE 474: Biochemical Engineering” as part of the Tomlinson Engagement Award for Mentoring (TEAM) program and developed sample problem sets and led five interactive review sessions with over 60 students attending each session

McGill University, Department of Biological and Biomedical Engineering **Sep 2020 - Dec 2020**
Teaching Assistant for BMDE 508

- Worked as teaching assistant for course “BMDE 508: Introduction to Micro and Nano-Bioengineering”.
- Updated and prepared the teaching modules and slides by identifying relevant material, preparing slides, and adding examples taken from the current literature.
- Conducted remote or in person sessions to answer student questions for preparation for the assignments, and helped prepare, manage, and conduct evaluations for assignments.

McGill University, Department of Biological and Biomedical Engineering **Sep 2021 - Dec 2021**
Grader for BMDE 503

- Worked as grader for course “BMDE 503: Biomedical Instrumentation”.
- Graded and managed evaluations for 9 assignments and answered student questions for the assignments on electronics and biomedical signal processing